

Computer Vision

Notes 2026

Introduction

- **Computer Vision** is the study and development of methods that enable the automation of tasks typically requiring human visual perception.

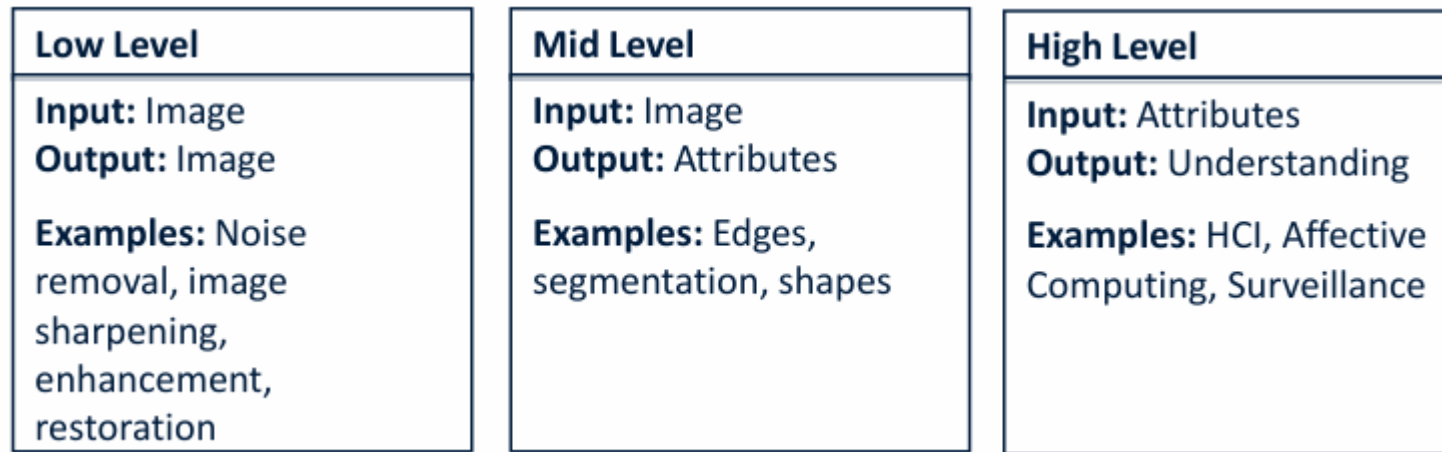


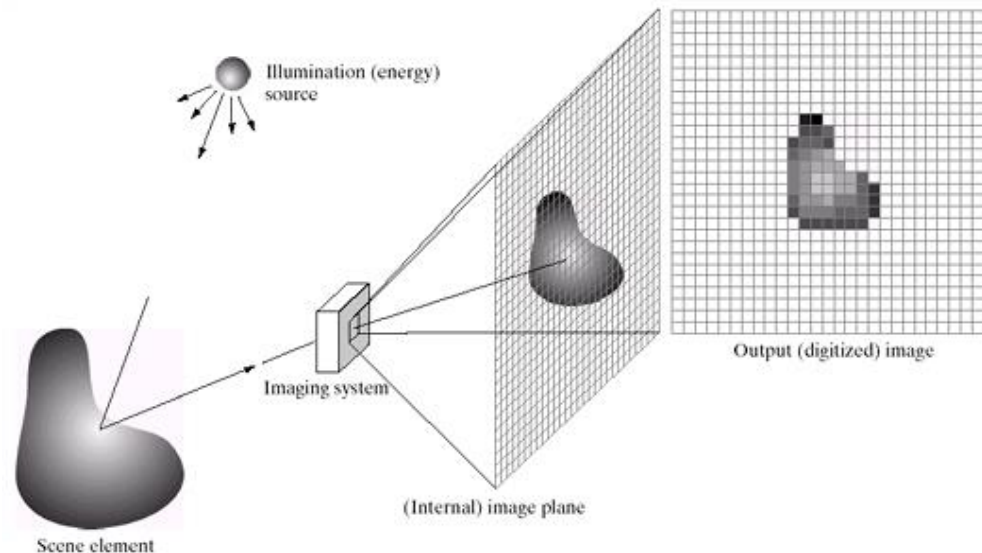
Image Processing  Computer Vision

1 - Foundations

Digital images, Edge detection, Morphology, Enhancement, Restoration

Digital Images

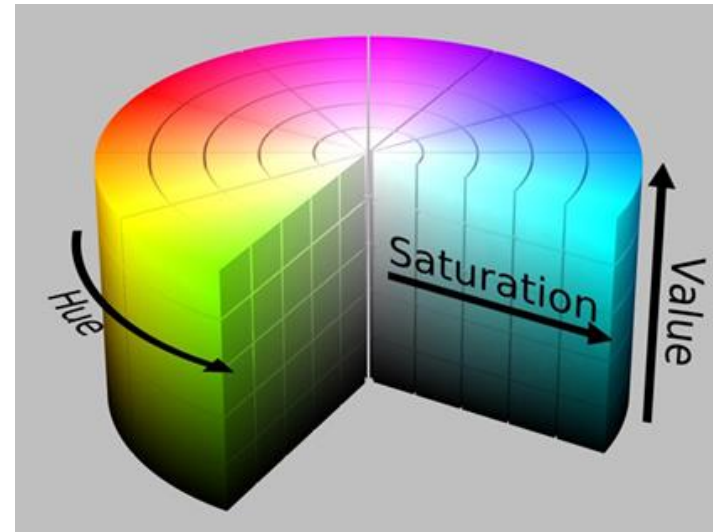
- A digital image is a 2D matrix of a finite set of **pixels**, encoding information about the *intensity* or *color* of light reflected by objects in a scene.
- The number of distinct colors that can be represented by a pixel depends on the number of bits per pixel (*bpp*).
- A digital image is a *coarse representation* of reality (quantization, etc.)



Digital Images - Colors

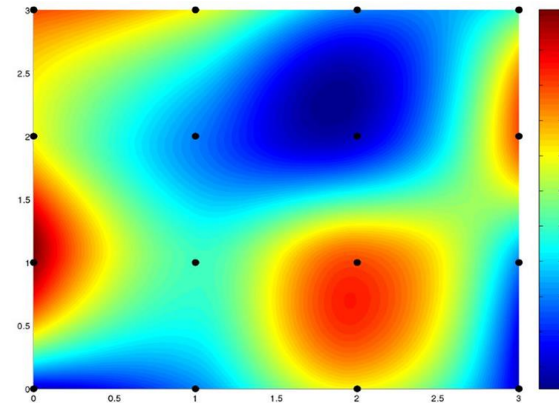
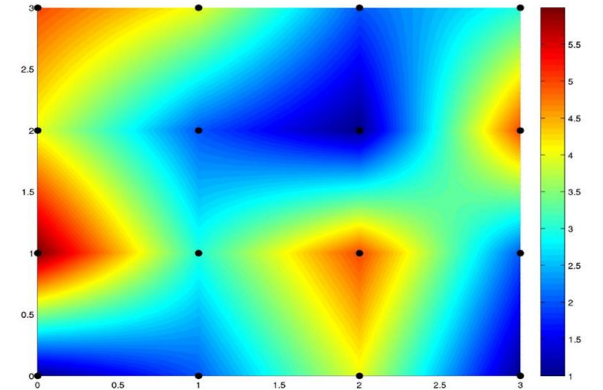
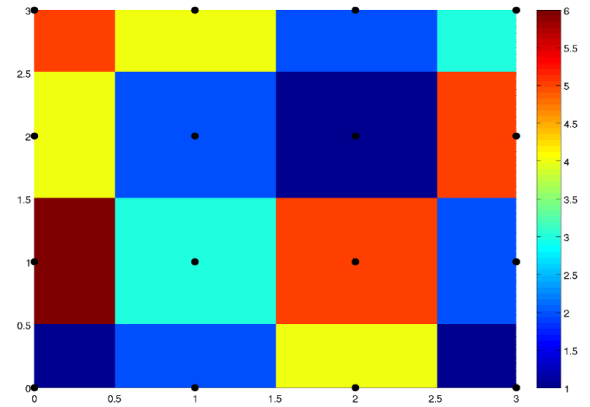
- There are 4 common types of colors used in images:
 - **Binary Images:** 1 bpp, either black or white.
 - **Grayscale:** 8 bpp, 256 gray levels (enough for recognition of most objects).
 - **RGB:** 3D vector for each pixel, used by most hardware.
 - **HSV:** 3D vector for each pixel, representing the **Hue** (spectral color), **Saturation** (intensity), and **Value** (luminescent contrast between black and white)

	0	40	80	120	160	200	255
Hue							
Saturation							
Luminance							



Digital Images - Interpolation

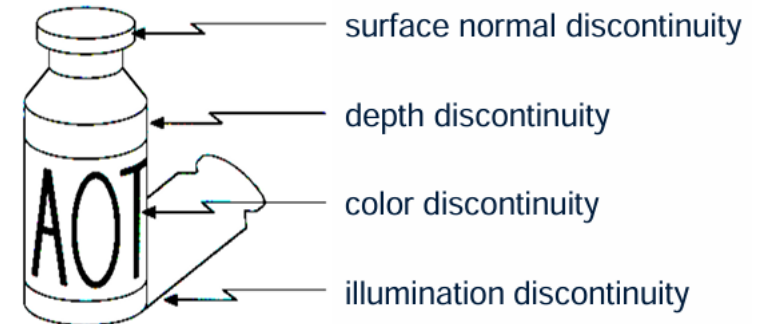
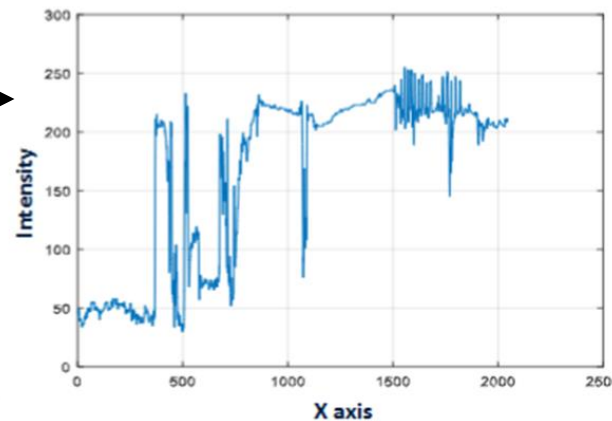
- Interpolation methods can be used to upscale images.
- **2D Nearest Neighbor** creates a piecewise function.
- **Bilinear Interpolation** approximates values using 4 known points.
- **Bicubic Interpolation** approximates value using 16 known points.





Edge Detection

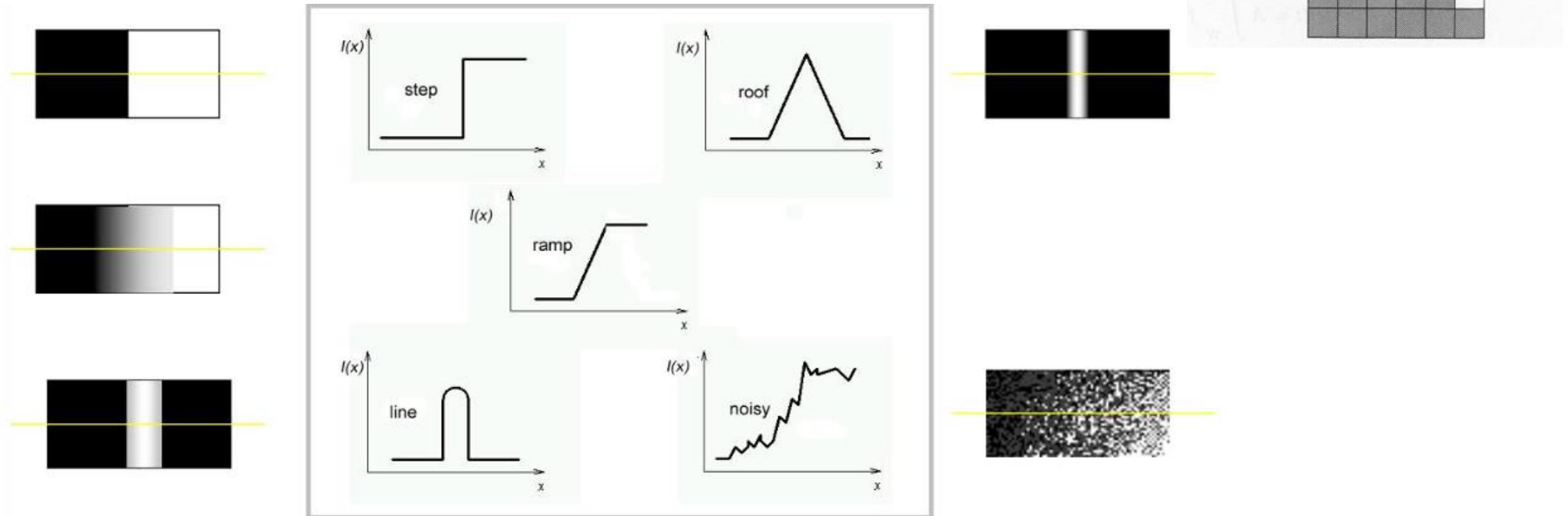
- Edges are significant *local* changes in intensity in an image.
- Intensity changes can also be caused by geometric events (surface changes, depth changes, color discontinuities, etc.) or other events (illumination changes, shadows, reflections, etc.).
- Important features can be extracted from edges (corners, lines, curves, etc.), making edge detection a common step in computer vision algorithms.





Edge Detection

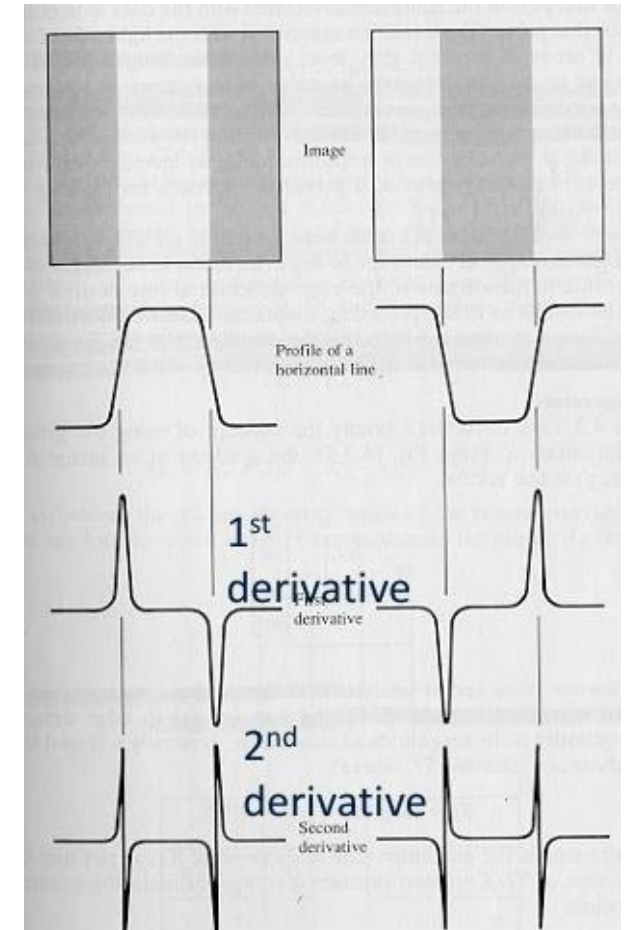
- The **Edge Direction** is perpendicular to the direction of maximum intensity change (the *edge normal*).
- There are multiple types of edges that can be detected:





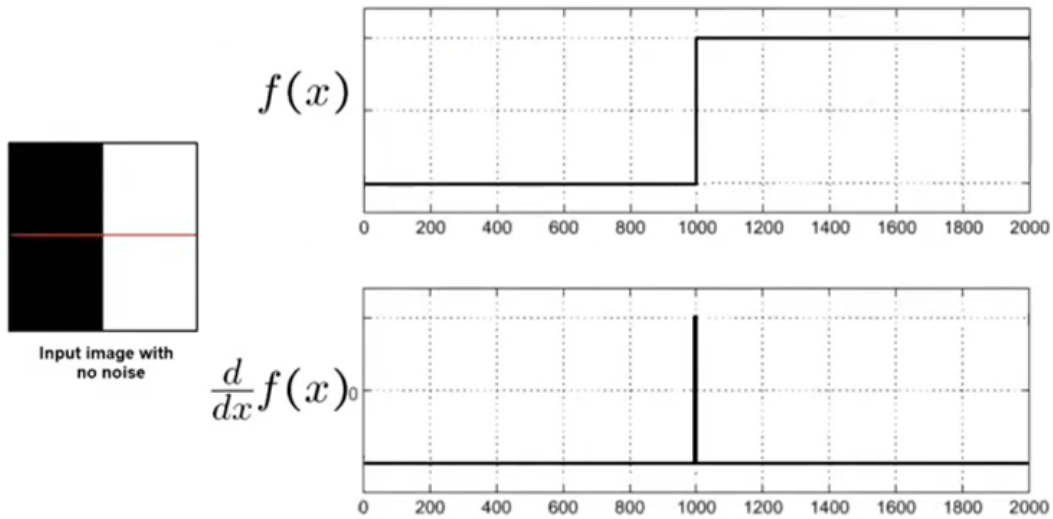
Edge Detection - Main Steps

1. **Smoothing:** suppress as much noise without destroying any true edges.
2. **Gradient Magnitude:** apply differentiation to enhance the quality of edges. Often done by:
 1. Detecting local extrema of the first derivative (requires setting a threshold).
 2. Detecting zero-crossings of the second derivative (more robust than first derivative).
3. **Thresholding:** determine which edges to retain and which to discard.
4. **Localization:** determine the exact edge locations.

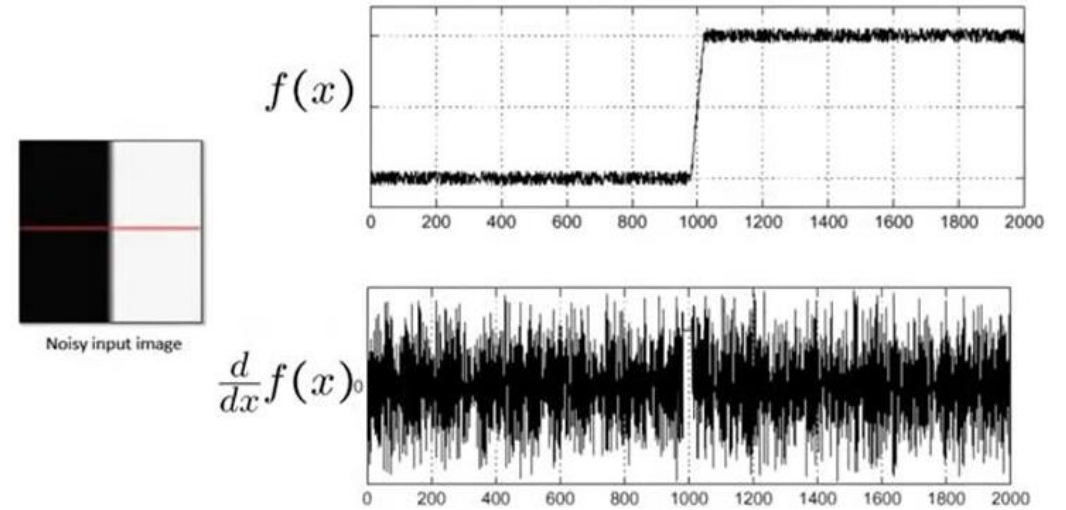




Edge Detection - First Derivative



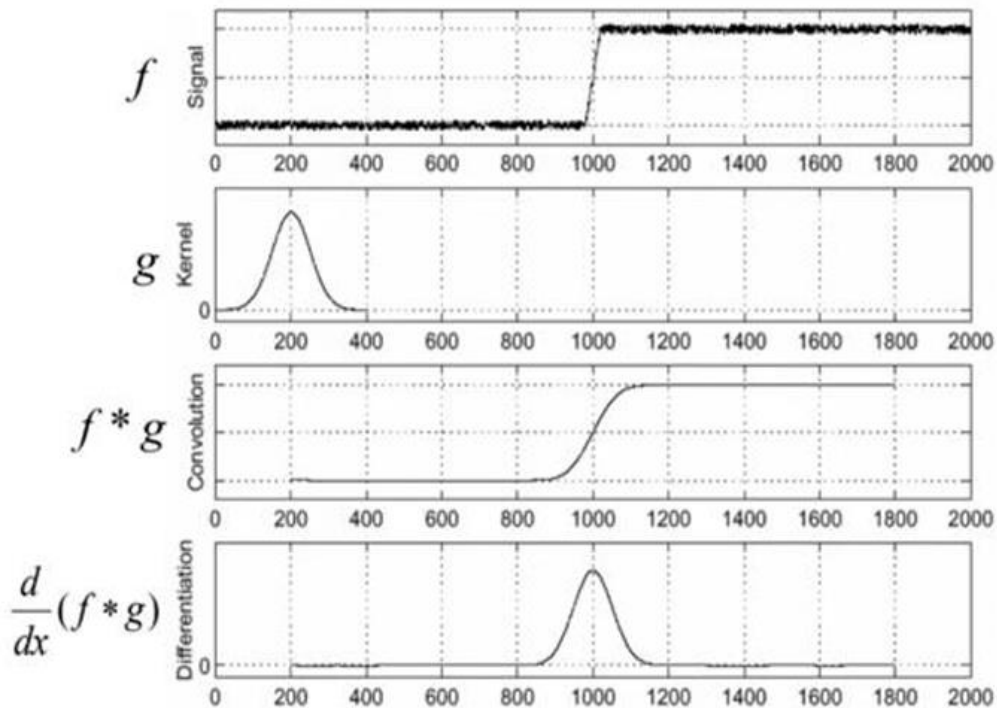
Applying the first derivative to an edge with no noise:
Clear spike that is easy to detect



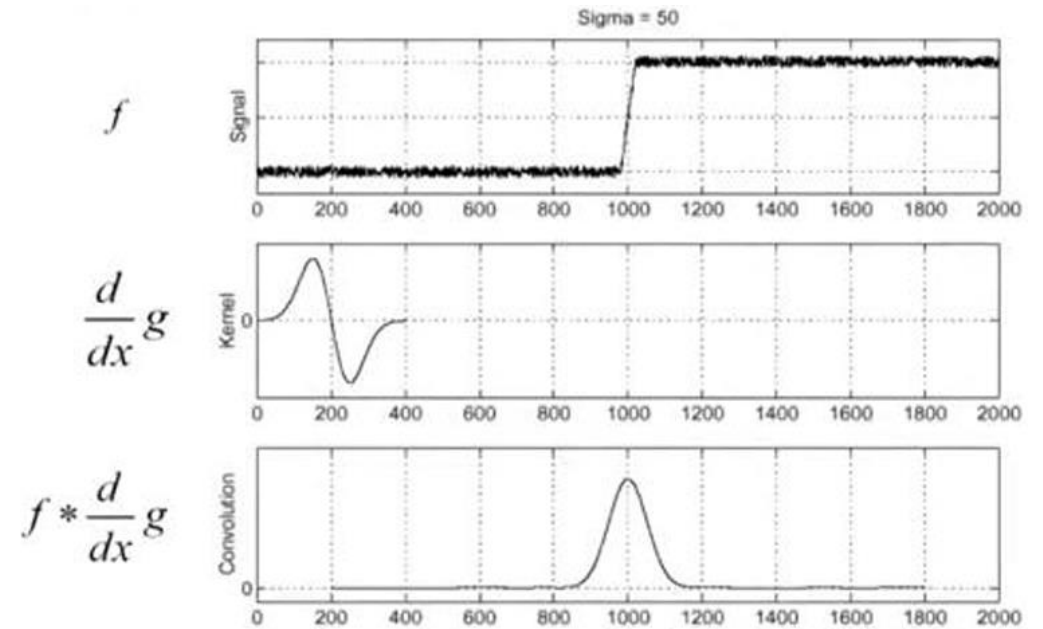
Applying the first derivative to an edge with noise:
Impossible to detect



Edge Detection - First Derivative



Solution: first apply a Gaussian filter to smooth the image

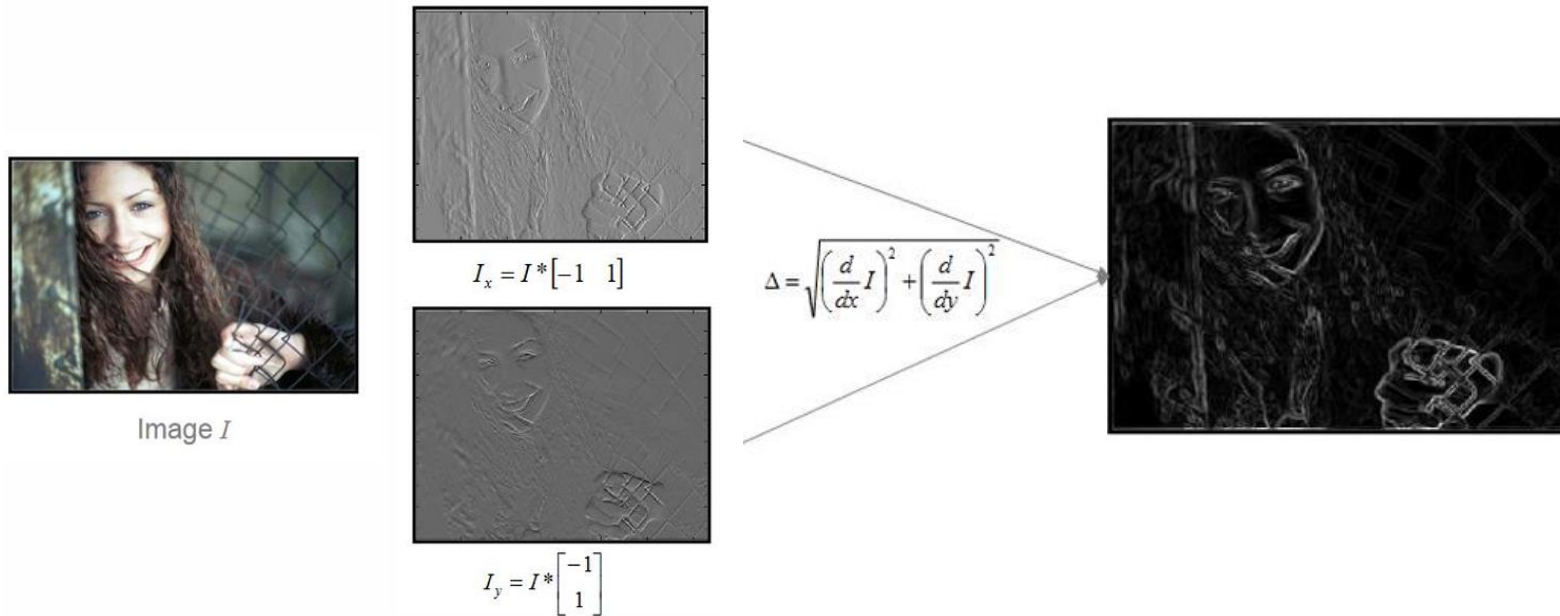


You can also apply the derivative of the Gaussian filter to skip having to differentiate the image



Edge Detection - First Derivative

- Since $\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h,y) - f(x,y)}{h}$, for $h = 1$ we get $\frac{\partial f}{\partial x} = f(x+1, y) - f(x, y)$.
- We can implement this approximation of $\frac{\partial f}{\partial x}$ as the mask $\begin{bmatrix} -1 & 1 \end{bmatrix}$.
- Similarly, we can implement an approximation of $\frac{\partial f}{\partial y}$ as the mask $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.



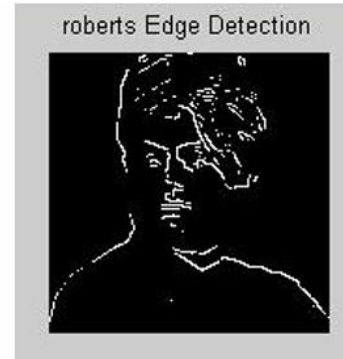
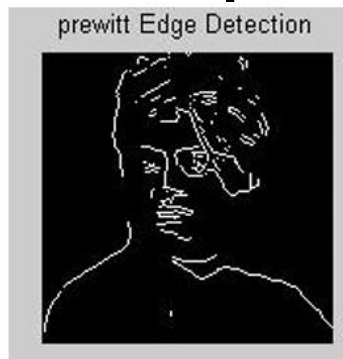


Edge Detection - First Derivative

- We can get better approximations by considering more neighboring pixels.
- **Robert's operator** considers 4 pixels: $\frac{\partial f}{\partial x} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $\frac{\partial f}{\partial y} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$
- Masks considering 9 pixels in a 3x3 neighborhood have a parameter c that gives emphasis to pixels closer to the center:

$$\frac{\partial f}{\partial x} = \begin{bmatrix} -1 & 0 & 1 \\ -c & 0 & c \\ -1 & 0 & 1 \end{bmatrix}, \frac{\partial f}{\partial y} = \begin{bmatrix} -1 & -c & -1 \\ 0 & 0 & 0 \\ 1 & c & 1 \end{bmatrix}$$

- For $c = 1$ this is known as the **Prewitt operator**, for $c = 2$ the **Sobel operator**.







Edge Detection - Second Derivative

- σ is the scale of the Gaussian filter:
 - Small σ detects fine features.
 - Large σ detects large edges.



original

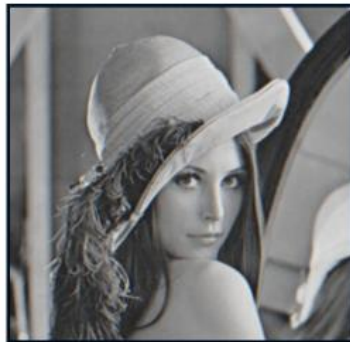


$\sigma = 1$

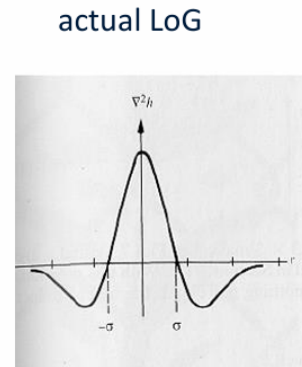
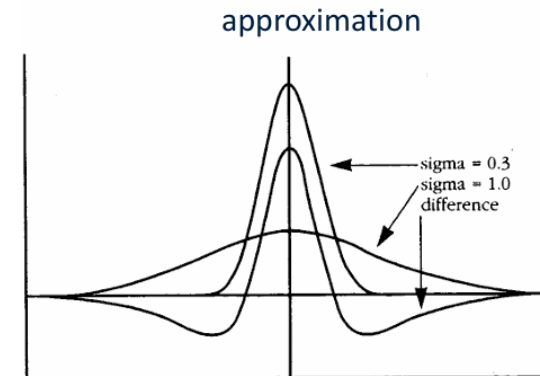


$\sigma = 2$

- The Laplacian of a Gaussian can also be approximated by the difference of two Gaussians functions (DoG): $\nabla^2 G \approx G(x, y; \sigma_1) - G(x, y; \sigma_2)$
- The best ratio of $\frac{\sigma_1}{\sigma_2}$ for this is around $\sqrt{2}$.

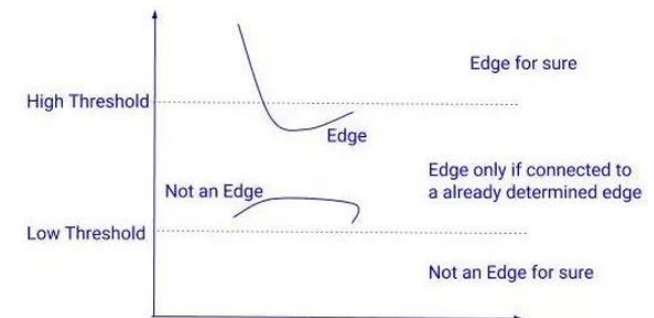


(b)-(a)



Edge Detection - Canny Edge Detection

- **Canny Edge Detection** is one of the most popular edge detection algorithms.
- It uses a multi-stage detection procedure:
 1. Smooth the image using a Gaussian filter.
 2. Detect edges using the Sobel operators.
 3. **Non-Maximum Suppression**: thin the edges by only keeping pixels that are local maxima in the gradient direction.
 4. **Double Thresholding**: apply a high- and low threshold: pixels above the high threshold are marked as *strong* edges, pixels below the high threshold but above the low threshold are marked as *weak* edges, pixels below the low threshold are removed.
 5. **Edge Tracking by Hysteresis**: weak edges are only kept if they are directly connected to a strong edge, other weak edges are removed.



Edge Detection - Canny Edge Detection



Original



Smoothed



Gradient magnitudes



Edges after
non-maximum suppression



Double thresholding



Edge tracking
by hysteresis

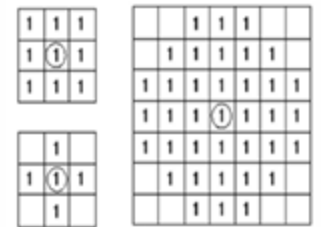


Final Output



Morphology

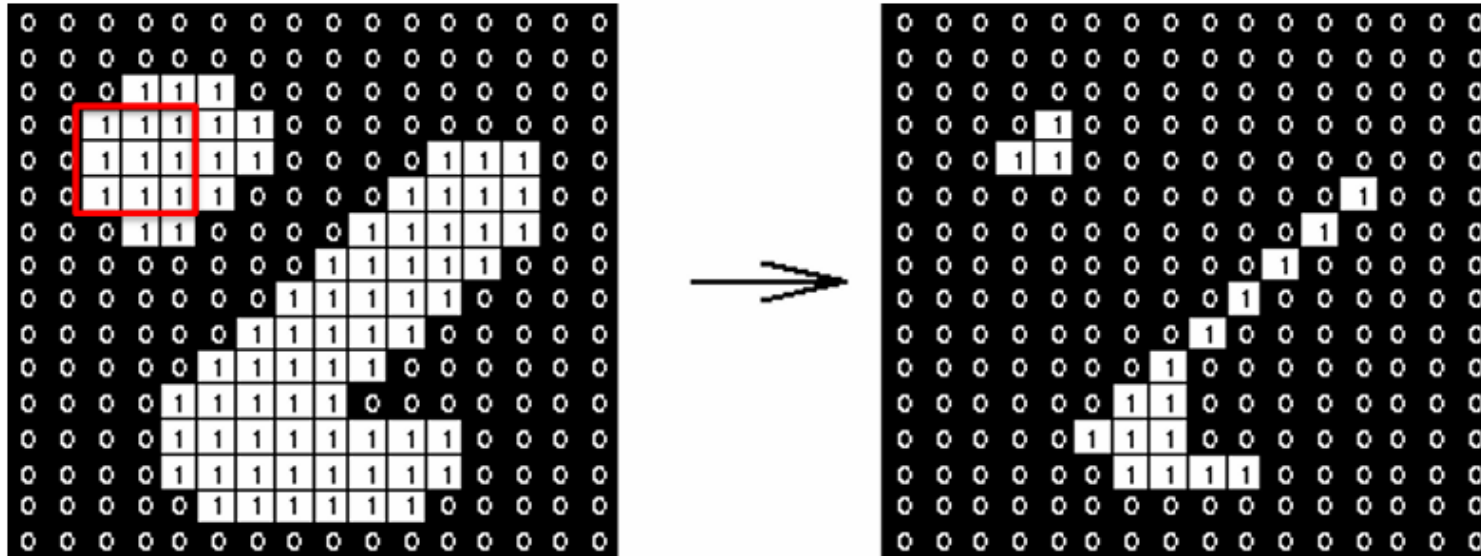
- **Morphological Image Processing** concerns a wide range of operators in image processing, mostly applicable to *binary images*.
- Images are probed with a small template, called the **Structuring Element** (SE). The structuring element is positioned at all possible locations in an image and is compared with the underlying set of pixels.
- An SE is a set of discrete points relative to an *origin*, often at the center of the pattern. In typical binary operations, points on the SE have a value of 1.





Morphology - Erosion

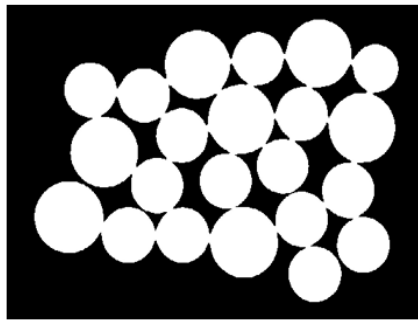
- **Erosion** is a fundamental morphological operation, reducing object size, eliminating small amounts of noise, and separating connected objects.
- $A \ominus B$: A pixel is set to 1 only if all pixels under the SE are 1 (*foreground*).
- The SE determines the precise effect of erosion on an image.



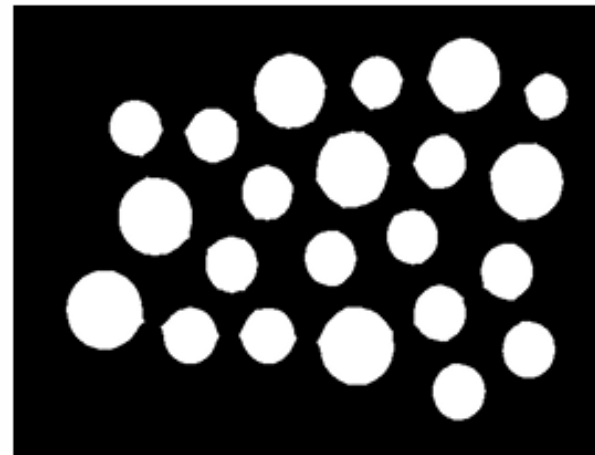


Morphology - Erosion

- The most common SE for erosion is the 3x3 square with origin at the center.
- Erosions can be made directional by using less symmetrical SEs, e.g.:
 - A 10x1 kernel will erode in the horizontal direction only.
 - A 3x3 square with origin in the top row will erode the bottom of an object more severely.
- Erosion can also be used to remove sporadic *salt and pepper noise*, and for edge detection by subtracting the eroded image from the original image.



Erosion with a
disk-shaped SE



The circles overlap in some places,
making it difficult to count them.

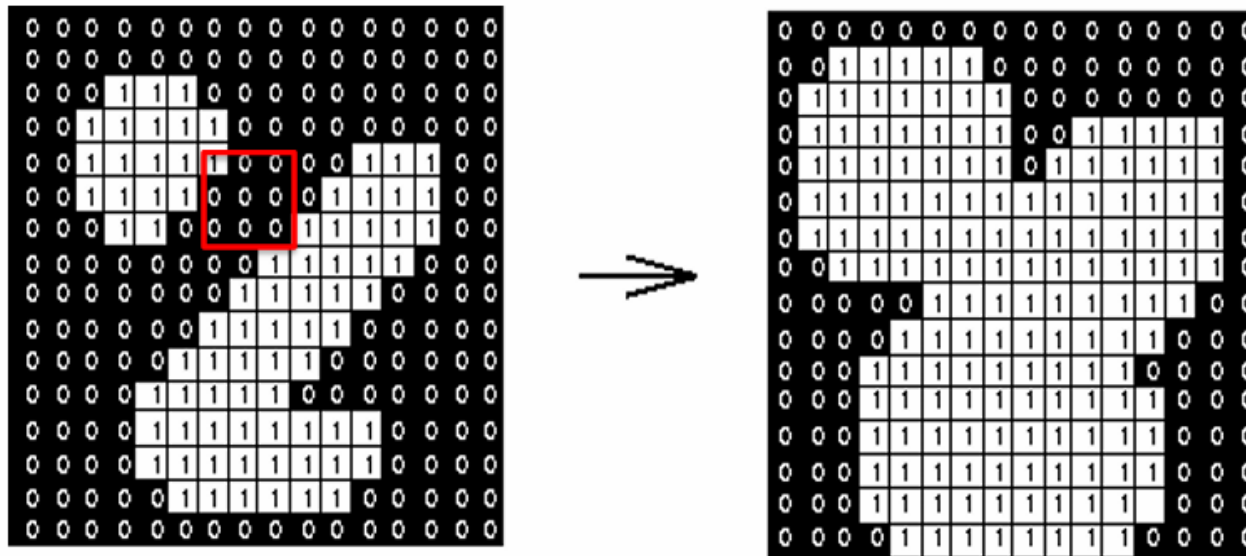
Circles have been separated,
while shape is preserved.

Note that SE *must* be smaller
than the smallest circle!



Morphology - Dilation

- **Dilation** is the *dual* operator to erosion, increasing object size, filling gaps, and connecting separated objects. Dilating foreground pixels is equivalent to eroding background pixels.
- $A \oplus B$: A pixel is set to 1 if at least one pixel under the SE is 1.
- The SE determines the precise effect of dilation on an image.





Morphology - Dilation

- With larger SEs, it is common to use disk-shaped SEs rather than squares.

- In general, when dilating with a disk-shaped SE:
 - Convex boundaries will become rounded.
 - Concave boundaries will be preserved.



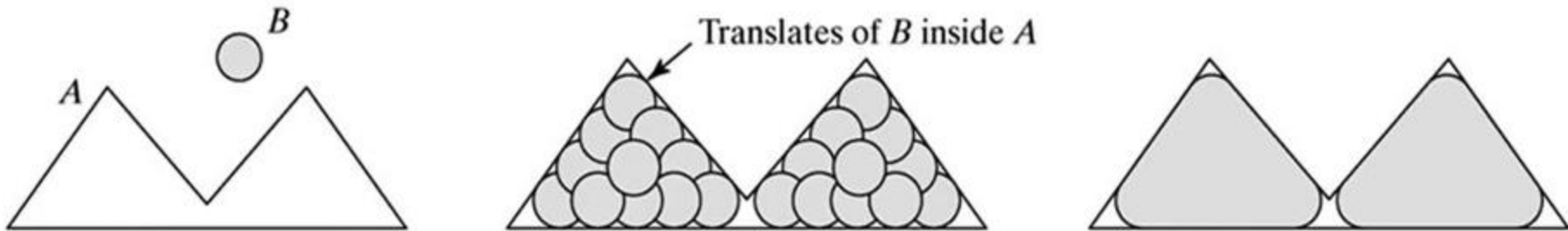
- Eroding with a disk-shaped SE has the opposite effect:
 - Convex boundaries will be preserved.
 - Concave boundaries will become rounded.





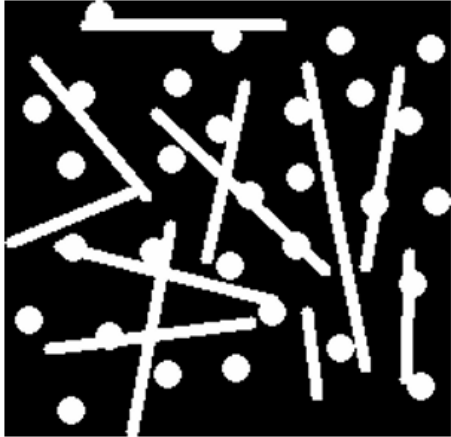
Morphology - Opening

- **Opening** is similar to erosion, removing foreground pixels near the edges of objects, while being less destructive. It is erosion, followed by dilation using the same SE: $A \circ B = (A \ominus B) \oplus B$
- The effect of the SE is to preserve foreground regions that have a similar shape to the SE, or that can completely contain the SE.
- For example, opening can be used to remove salt and pepper noise without affecting any other regions of foreground pixels.

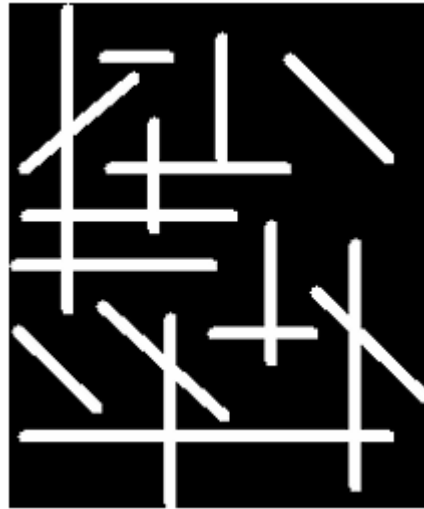
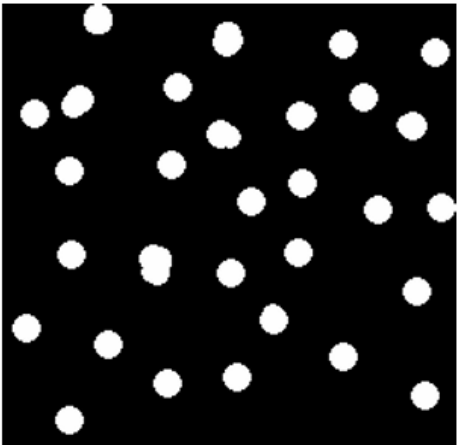




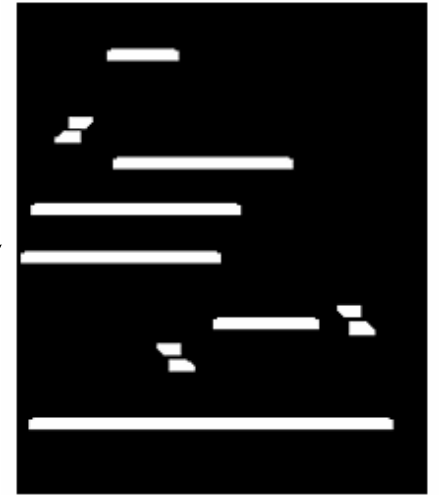
Morphology - Opening



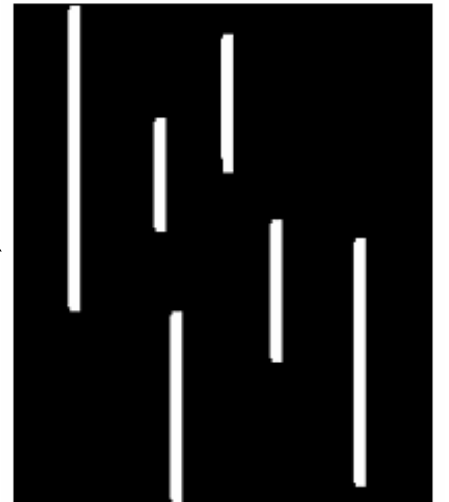
Opening with a circle SE
(diameter = 11 pixels)



Opening with a 9x3 SE



Opening with 3x9 SE



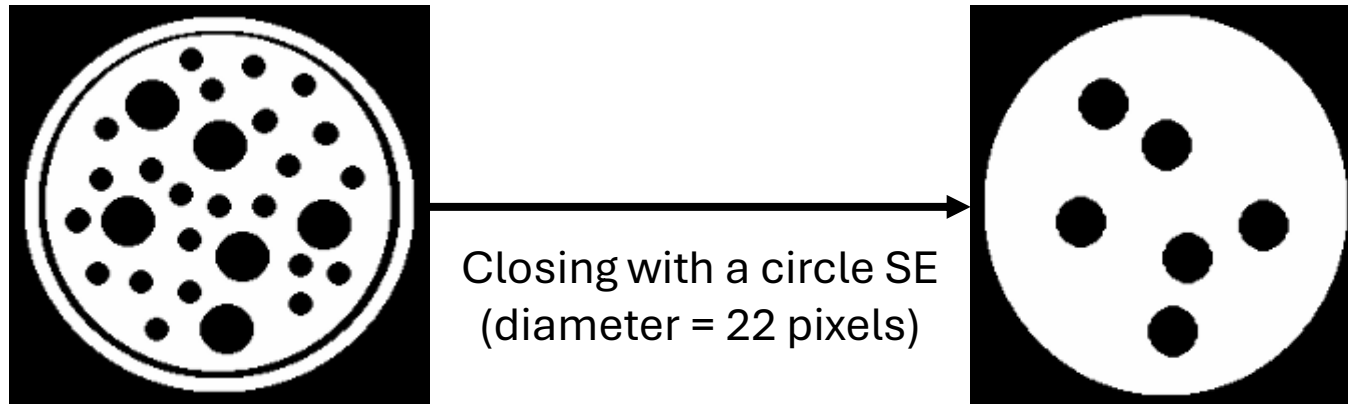
All lines removed with only a few
elements distorted (erosion would
destroy a lot more circles!)

Downside: we must know the exact shape / size beforehand!)



Morphology - Closing

- **Closing** is similar to dilation, enlarging foreground pixels near the edges of objects, while being less destructive. It is dilation, followed by erosion using the same SE: $A \cdot B = (A \oplus B) \ominus B$
- The effect of the SE is to preserve foreground regions that have a similar shape to the SE, or that can completely contain the SE.
- For example, closing can be used to fill in salt and pepper noise without affecting any other regions of foreground pixels.



Morphology - Opening & Closing

Property	Notation
Opening is <i>idempotent</i>	$(A \circ B) \circ B = A \circ B$
Opening is <i>increasing</i>	If $A \subseteq C$, then $A \circ B \subseteq C \circ B$
Opening is <i>anti-extensive</i>	$A \circ B \subseteq A$
Opening is <i>translation-invariant</i>	$A_x \circ B = (A \circ B)_x$
Closing is <i>idempotent</i>	$(A \cdot B) \cdot B = A \cdot B$
Closing is <i>increasing</i>	If $A \subseteq C$, then $A \cdot B \subseteq C \cdot B$
Closing is <i>extensive</i>	$A \subseteq A \cdot B$
Closing is <i>translation-invariant</i>	$A_x \cdot B = (A \cdot B)_x$



Image Enhancement

- Not all images are perfectly distributed, as can be seen by its *histogram*:

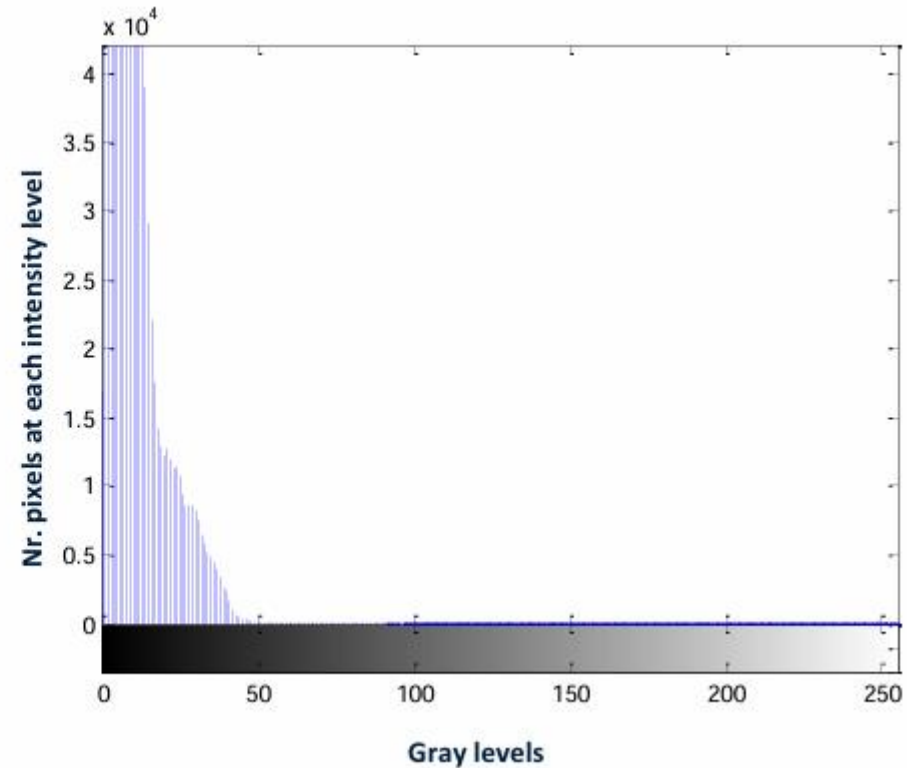




Image Enhancement - Contrast Stretching

- **Contrast Stretching** is a linear enhancement technique that stretches an image's contrast to fill the full dynamic range (0-255):
$$s = \frac{(r - r_{min})}{(r_{max} - r_{min})} \cdot 255$$
- s is the output pixel intensity, r is the input pixel intensity, and r_{min}, r_{max} are the lowest and highest pixel intensities in the original image respectively (more commonly the 5th and 95th percentile for more robust results).

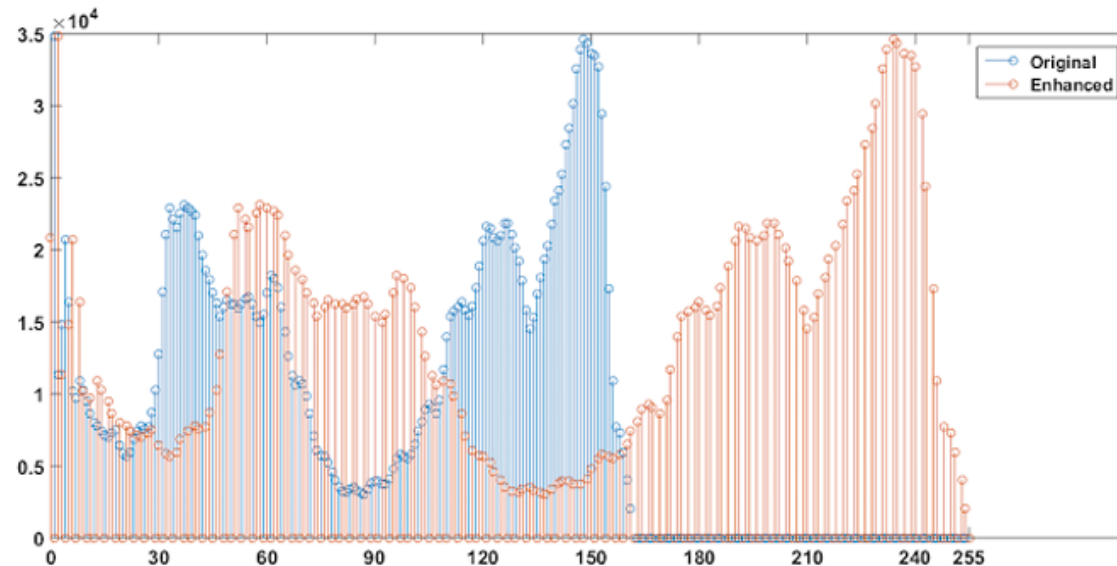
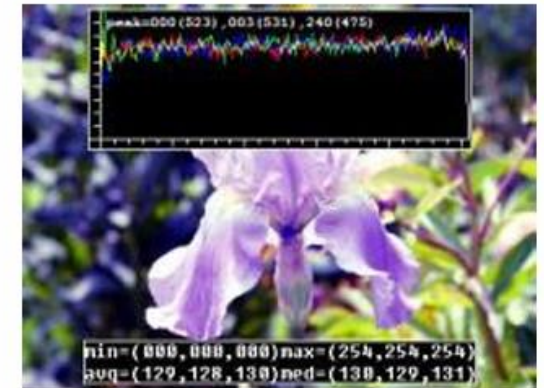
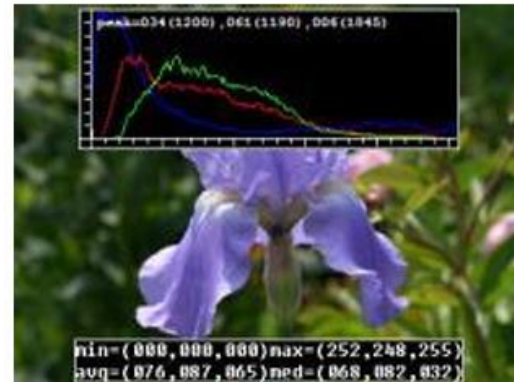




Image Enhancement - Histogram Equalization

(permanent loss of data!)

- **Histogram Equalization** is an enhancement technique that uniformly distributes pixel intensities across the full dynamic range (0-255).
- Assuming gray level i occurs n_i times in the image, and the total number of pixels is $n_0 + n_1 + \dots + n_{255} = n$, histogram equalization tries to improve the image by changing gray level i to $\left(\frac{n_0 + n_1 + \dots + n_i}{n}\right) \cdot 255$, rounded to the nearest integer (in other words, scale the PDF (histogram) according to the CDF).



Can't be used with color images! (new colors appear)



Image Enhancement - Spatial Filtering

- **Spatial Filtering** (also called **Convolution**) is the process of sliding a *filter*, *mask*, or *kernel* over an image.
- Convolution is *associative*, meaning: $F * (G * I) = (F * G) * I$
- Some example filters:

3x3 moving average filter

$\frac{1}{9}$	1	1	1
	1	1	1
	1	1	1

3x3 sharpening filter

$\frac{1}{9}$	-1	-1	-1
	-1	8	-1
	-1	-1	-1

Sobel operators

-1	0	1
-2	0	2
-1	0	1

to compute $\frac{\partial P}{\partial x}$

-1	-2	-1
0	0	0
1	2	1

to compute $\frac{\partial P}{\partial y}$

Shift right by 1 pixel

0	0	0
1	0	0
0	0	0

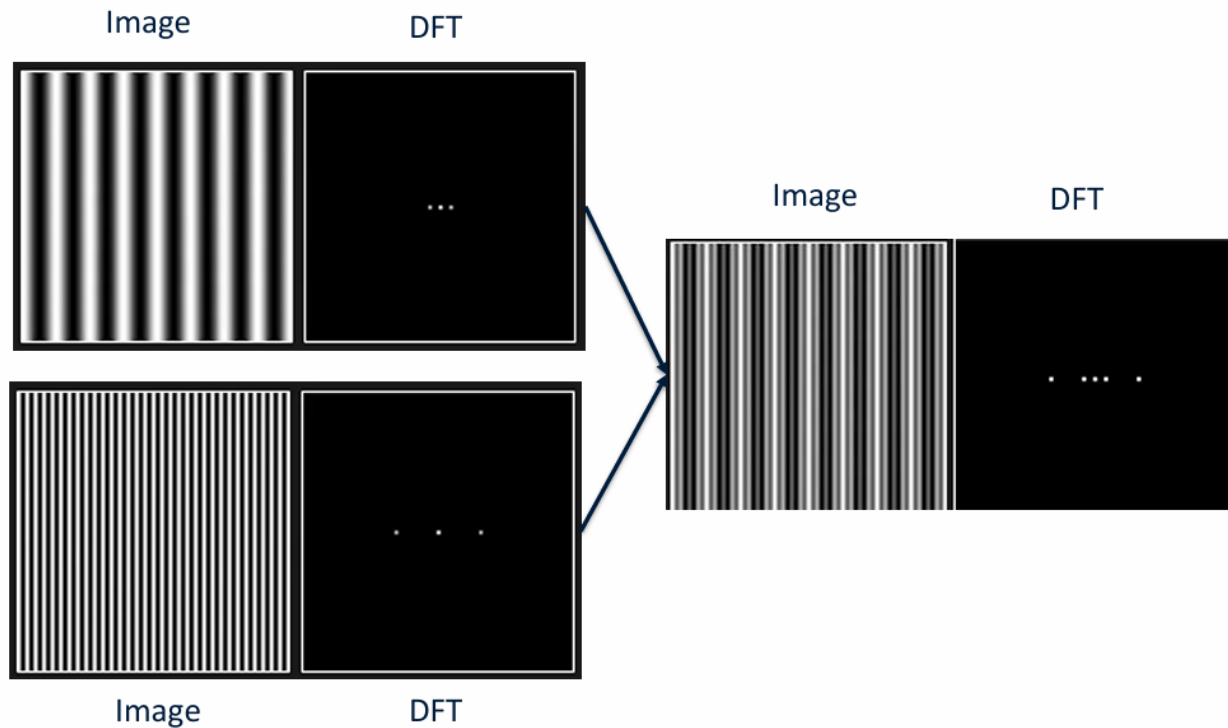


Image Enhancement - Spatial Filtering

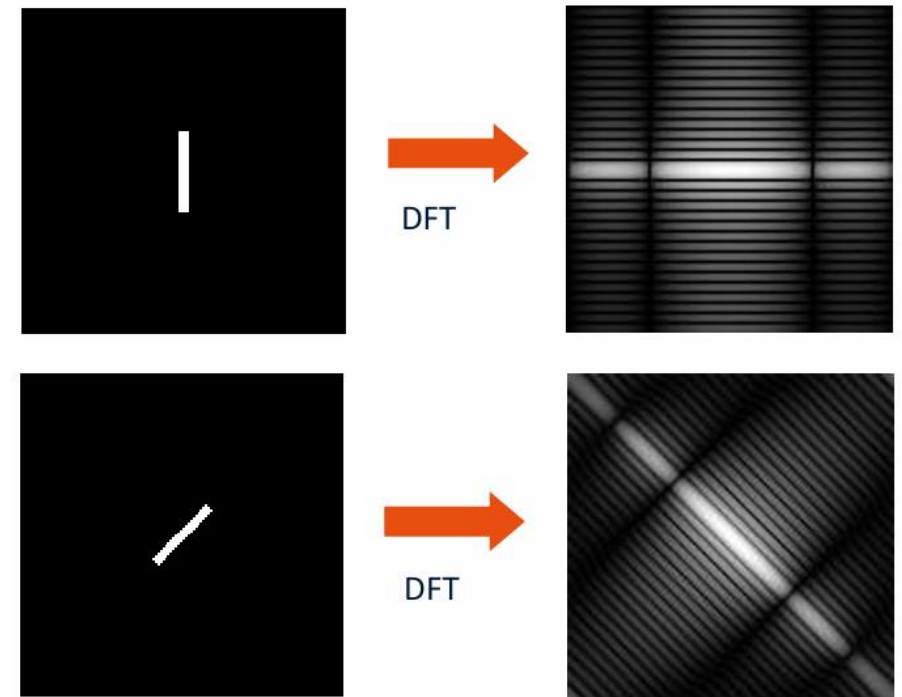
- Filters can also be applied in the *frequency domain*. To get there, you must apply a 2D **Discrete Fourier Transform (DFT)** on the image.
- The 2D DFT is given as $F(u, v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-2\pi i (\frac{ux}{M} + \frac{vy}{N})}$, where u, v are frequency variables and M, N are the dimensions of the image.
- The inverse 2D DFT is given as $f(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{2\pi i (\frac{ux}{M} + \frac{vy}{N})}$
- Amongst others, the 2D DFT is:
 - *Symmetric*: $|F(u, v)| = |F(-u, -v)|$
 - *Scaling*: $aF(u, v) \Leftrightarrow af(x, y)$
 - *Periodic*: $F(u, v) = F(u + M, v) = F(u, v + N) = F(u + M, v + N)$
 - *Separable*: equal to 1D DFT in one dimension, followed by 1D DFT in the other



Image Enhancement - Spatial Filtering



The frequency spectrum is often visualized through a *magnitude plot*

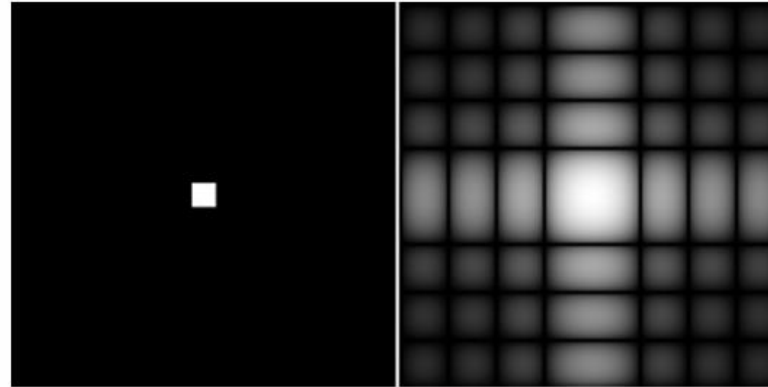


Edges in the original image are visible on the magnitude plot at a 90° angle to the original line

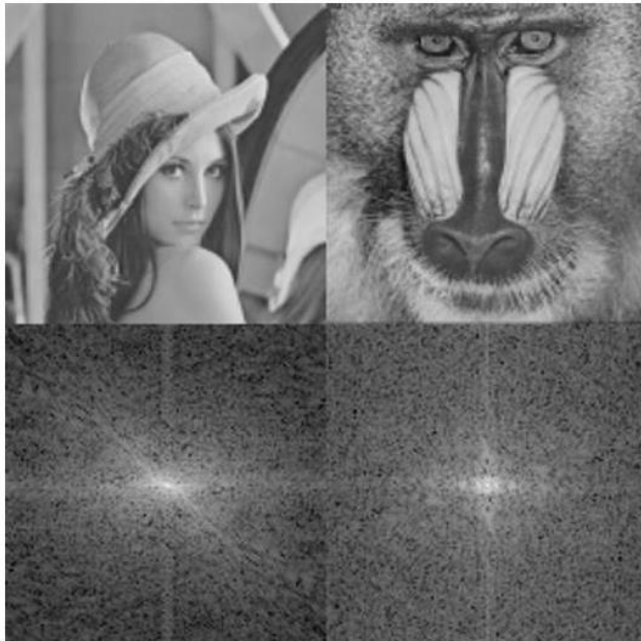


Image Enhancement - Spatial Filtering

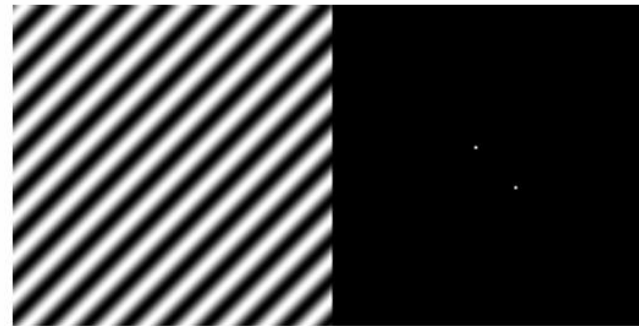
Rectangle



Its DFT



Sine wave



Its DFT

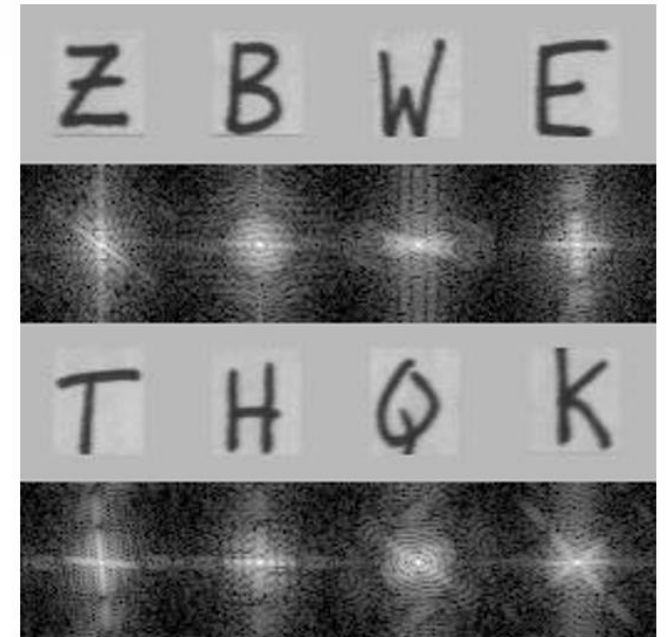
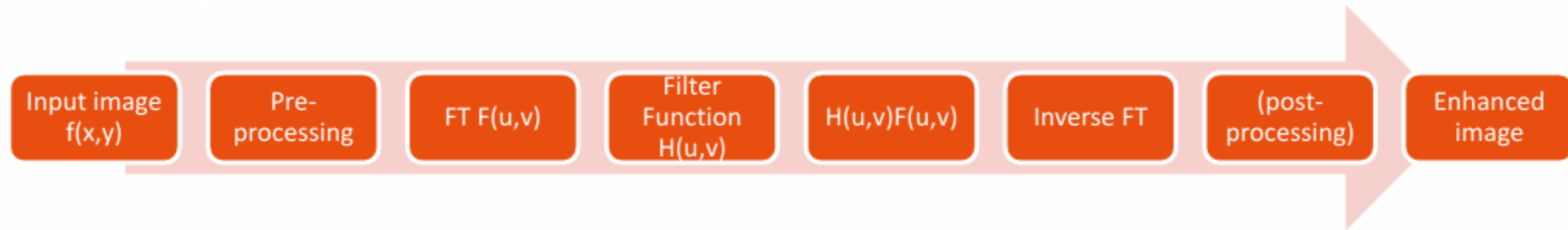




Image Enhancement - Spatial Filtering

- Convolution in the *image domain* is equivalent to multiplication in the *frequency domain* (and the other way around).
- We can therefore also apply filters the following way:

Note that both the image and filter need to be 0-padded to a size of at least $A+C-1$ to make multiplication possible



- One of the most basic filters is the **Notch Filter**: forcing the average value of the image to 0.

$$H(u, v) = \begin{cases} 0 & \text{if } (u, v) = (0,0) \\ 1 & \text{otherwise.} \end{cases}$$



Image Enhancement - Spatial Filtering

- Other important filters include the **Low-Pass Filter (LPF)** and the **High-Pass Filter (HPF)**.
- The simplest LPF is the **Ideal Low-Pass Filter**, cutting off high-frequency components further away from the origin than a specified distance D_0 :

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases}$$

- Where $D(u, v)$ is the distance of (u, v) to the center of the magnitude plot.

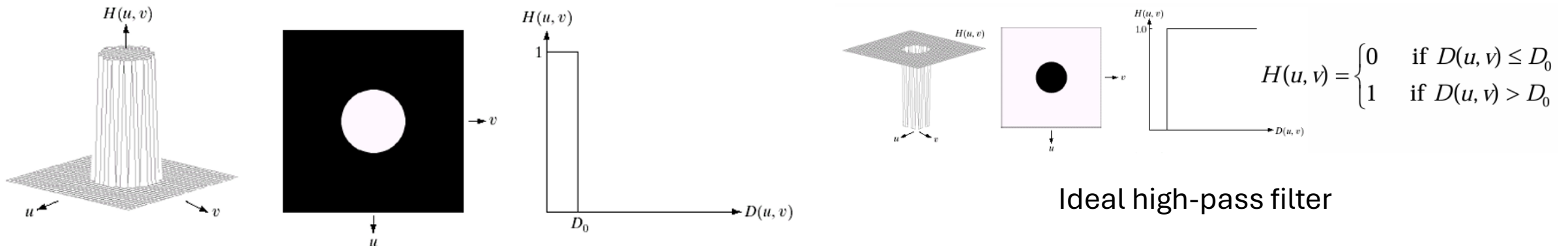
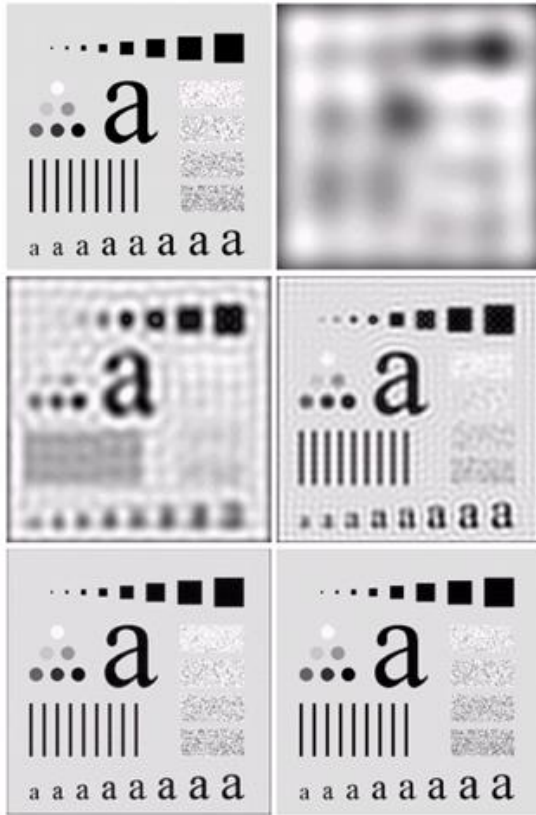
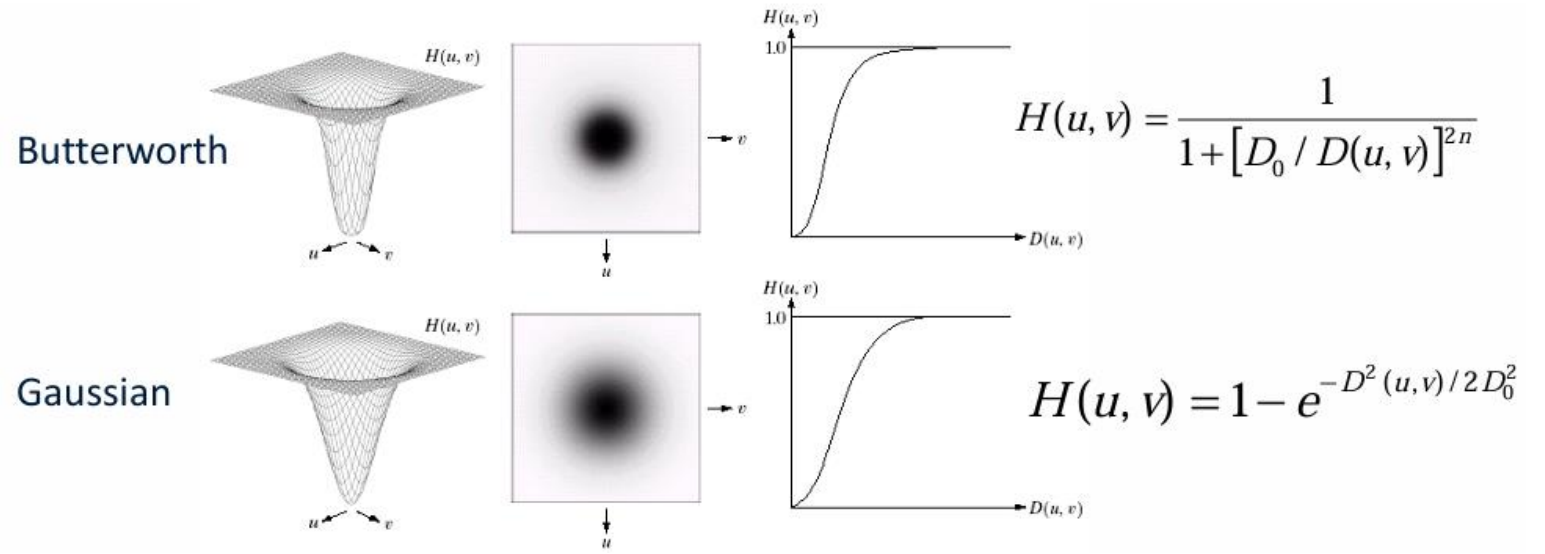




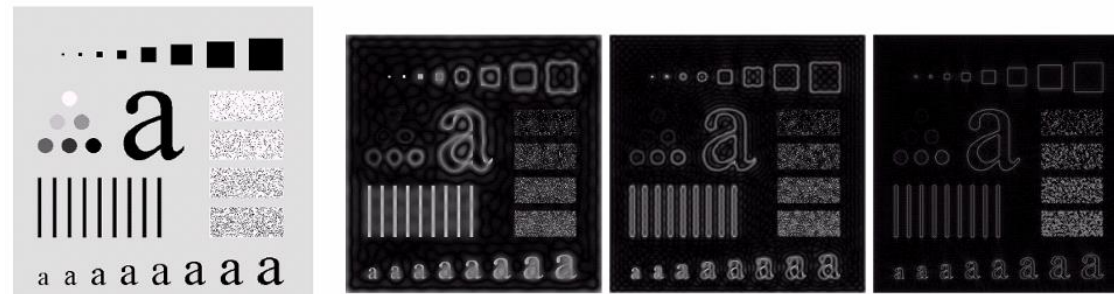
Image Enhancement - Spatial Filtering



Applying an ideal LPF with various cut off frequencies. *Smoothing* is fundamentally applying a LPF in the frequency domain.



Other variants include the **Butterworth** L/HPF and the **Gaussian** L/HPF (here shown as HPFs)

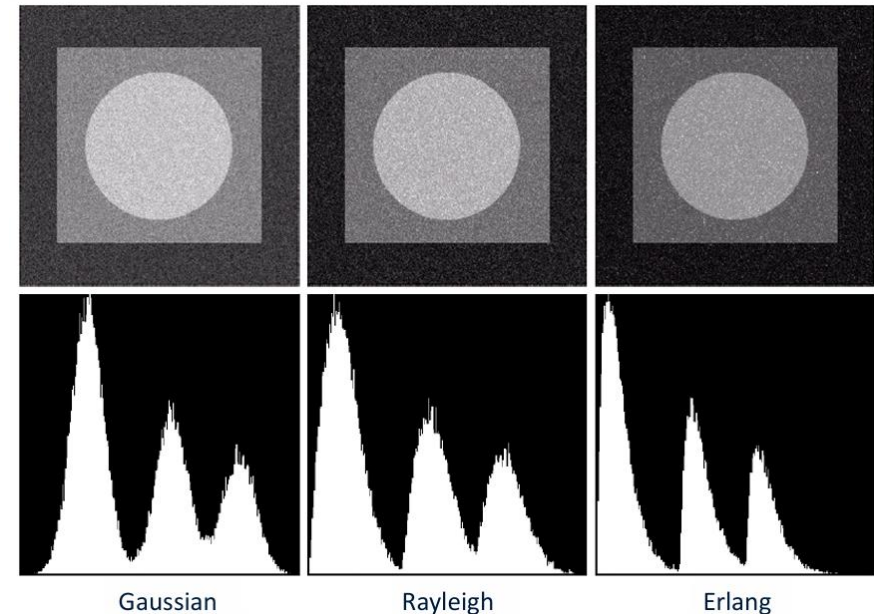
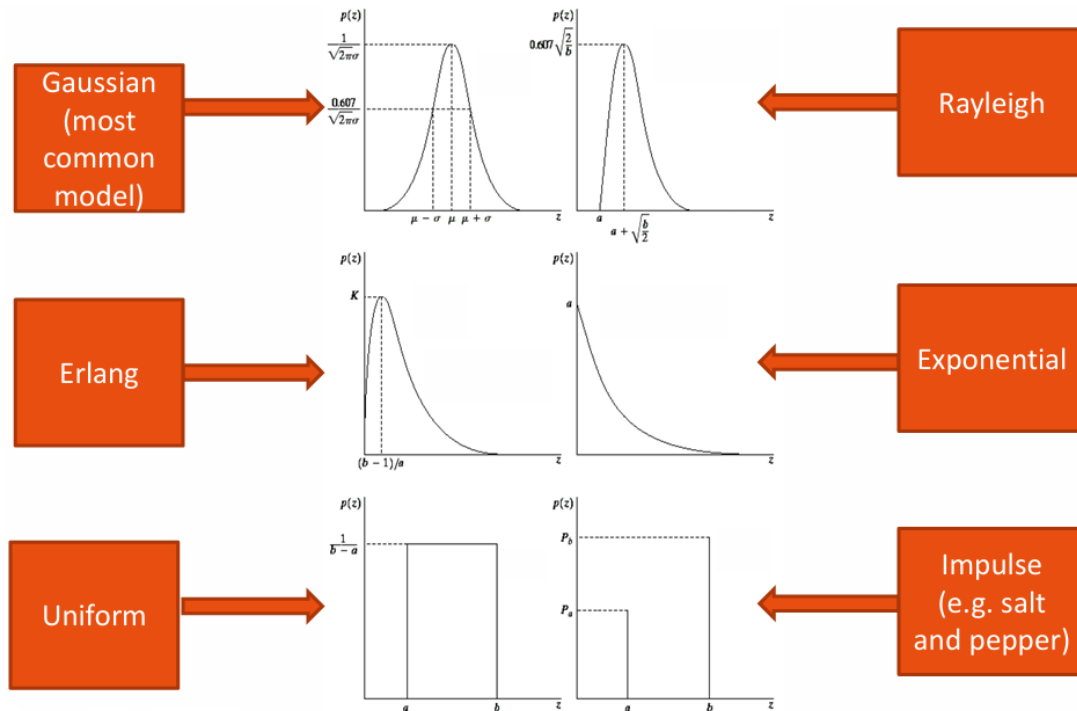


Applying an ideal HPF with various cut off frequencies.



Image Restoration

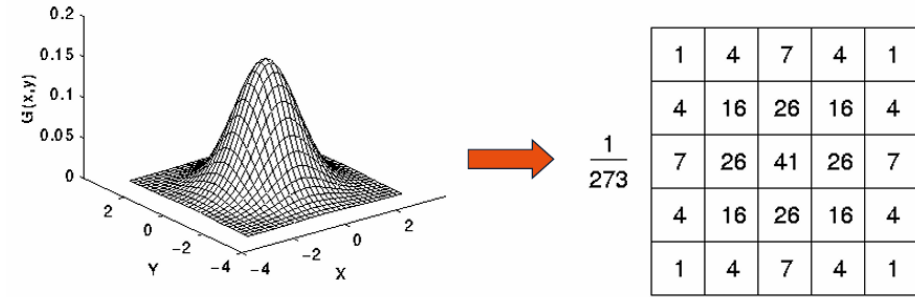
- **Image Restoration** concerns the removal of unwanted *noise* on an image.
- A noisy image can be modelled as $g(x, y) = f(x, y) + \eta(x, y)$
- Having an estimate of the noise model η can greatly help with restoration.



Different noise models lead to different histograms



Image Restoration



- Most noise can be removed through blurring, with the most common techniques being a *moving average filter* or a *Gaussian filter*.
- Sometimes noise is periodic, in which case it can best be removed in the frequency domain using a **Band Reject Filter**.

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) < D_0 - \frac{W}{2} \\ 0 & \text{if } D_0 - \frac{W}{2} \leq D(u, v) \leq D_0 + \frac{W}{2} \\ 1 & \text{if } D(u, v) > D_0 + \frac{W}{2} \end{cases}$$

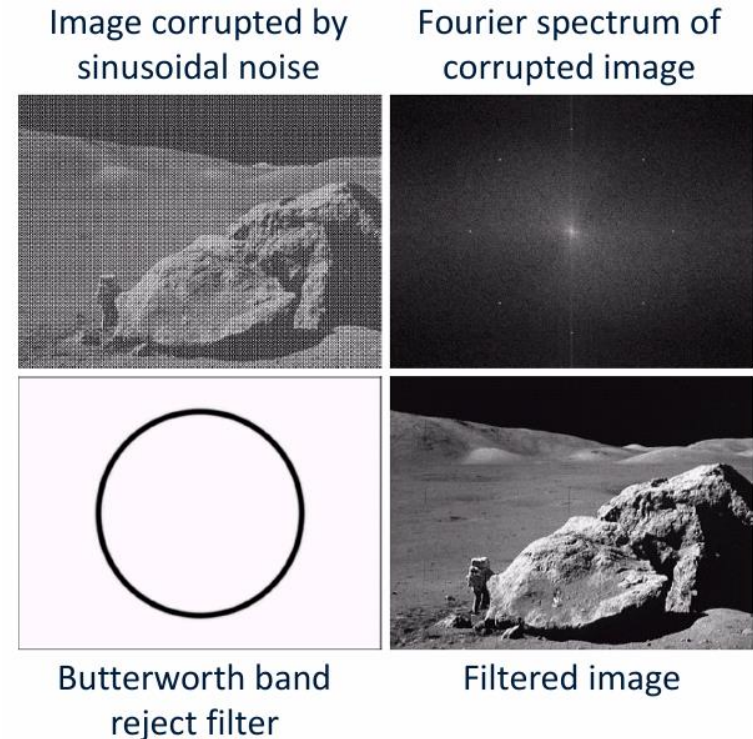
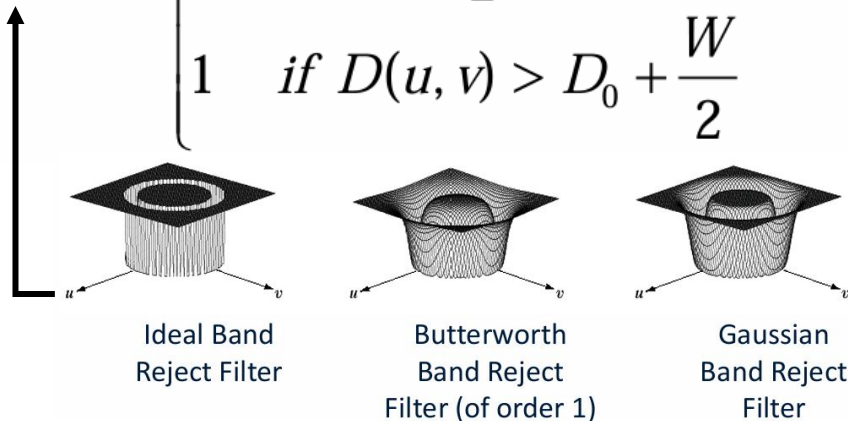




Image Restoration - Inverse Filtering

- Applying a filter in the frequency domain by multiplying the DFTs of the image and the filter is known as the **Convolution theorem**: $Y(u, v) = X(u, v)H(u, v)$
- If we have a filtered image Y and a filter H , we can recover the original image X by dividing $\frac{Y}{H}$.



$$X(u, v) = \begin{cases} \frac{Y(u, v)}{H(u, v)}, & \text{if } |H(u, v)| > d \\ Y(u, v), & \text{otherwise} \end{cases}$$

Constrained Division with threshold d



Image with an LPF applied